

Dhuruv Kumar Gupta

Python Developer | AI/ML & Generative AI

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PROFESSIONAL SUMMARY

Computer Science and AI/ML undergraduate (**2027 batch**) with a strong foundation in **Python programming, Data Structures & Algorithms, and problem-solving**, backed by **2 software development internships** and independent projects in **Generative AI, LLMs, RAG, and AI agents**. Hands-on experience building Python applications with FastAPI, vector embeddings, and semantic search, alongside full-stack development using React, Node.js, and PostgreSQL.

TECHNICAL SKILLS

Programming & Problem-Solving: Python, C++, SQL, Data Structures & Algorithms

AI/ML & Generative AI: Artificial Intelligence, Machine Learning, Deep Learning, Generative AI, LLMs, RAG

Backend & APIs: FastAPI, Node.js, Express.js, RESTful APIs

Frontend: React.js, HTML, CSS, JavaScript

Databases: PostgreSQL, MongoDB, MySQL, Supabase, DBMS

Cloud & Tools: AWS, Render, Vercel, Git, GitHub

PROJECTS

AI Research Assistant – Python, FastAPI, RAG, LLMs

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- Built a **Retrieval-Augmented Generation (RAG)** application in Python (FastAPI, ChromaDB, vector embeddings, semantic search) exposed through a REST API architecture.
- Implemented Python-based pipelines for document ingestion, document chunking, embedding generation, and vector retrieval, reaching **90% relevance accuracy** on test queries.
- Designed API endpoints for authentication, semantic search, and AI-powered retrieval, applying core data-structure and algorithmic thinking to optimize query performance.

ShortsLab – AI Shorts Generator (Python, LLMs, AI Automation)

Live | GitHub

- Developed a Python-based AI video-processing pipeline (FastAPI, Faster Whisper, Gemini LLM) that automates long-form video repurposing into short-form clips.
- Used the Gemini LLM to automatically identify engaging video segments from timestamped transcripts.
- Automated the pipeline into **4 sequential stages** (upload, transcription, timestamp selection, and short-video generation) using FFmpeg and MoviePy, eliminating manual video editing from the workflow.

Budget Rectifier Platform

Live | GitHub

- Architected a finance application (Next.js, TypeScript, Supabase, PostgreSQL) for expense and budget management with an indexed, transaction-safe schema design.
- Shipped **6+ core features**, including expense tracking, monthly budgets, category analytics, secure authentication, transaction history, and dashboard insights, adopted by **20+ test users** during development.
- Analyzed transaction data across categories to surface spending patterns, improving budget-adherence visibility through real-time dashboard insights.

EXPERIENCE

Software Developer Intern, Varach Innovations

Jun 2026 – Present

- Engineered backend services and API endpoint in Node.js and Supabase, applying clean architecture, async/await-based logic, and structured error handling across production workflows.
- Delivered 4 core workflows for EventPass, a production event-management platform (React, Vite, Tailwind CSS, Node.js, Supabase, PostgreSQL), covering event creation, participant registration, ticketing, and attendance management.
- Digitized the ticketing and attendance workflow, replacing manual, paper-based check-in with a single streamlined process used across all EventPass events.
- Partnered with a cross-functional team of 3 across backend, frontend, and database development, contributing to testing, debugging, and deployment cycles using Git/GitHub for version control and code review.

Web Developer Intern, CodSoft

Dec 2025 – Jan 2026

- Built a web-based job portal with **JWT-based authentication**, role-based authorization across **2 user roles** (recruiter and candidate), candidate management, and application-tracking workflows.
- Designed and shipped proper HTTP methods and status codes supporting job posting, authentication, candidate information, and application processing.
- Refactored backend component architecture and data organization into modular, reusable services, improving maintainability and accelerating new-feature development.

EDUCATION

Uttarakhand Technical University, Dehradun

2023 – 2027

B.Tech in Computer Science and Engineering (AI & ML) | CGPA: **7.57/10**

CERTIFICATIONS

- Generative AI** – Physics Wallah & Microsoft [Credential]
- RAG with MongoDB** – MongoDB [Credential]
- AI Agent with MongoDB** – MongoDB [Credential]
- Vector Search Fundamentals** – MongoDB [Credential]
- AI Data Strategy** – MongoDB [Credential]
- Basics of MongoDB** – MongoDB [Credential]